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**Learning distance-dependent spatial structure for multichannel
source separation**

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Distance between sound sources and a microphone array fundamentally shapes acoustic observations through level attenuation, spectral coloration, direct-to-reverberant ratio (DRR), and spatial coherence. However, many multichannel source separation methods treat sources primarily as abstract signals and do not explicitly model distance-dependent structure, particularly in reverberant environments. This paper investigates whether distance-dependent spatial structure can be learned and exploited to improve multichannel source separation. We propose a distance-aware framework consisting of two stages. First, a neural distance embedding network (DistanceNet) is trained using multichannel short-time Fourier transform features to regress source-to-array distance, classify sources as near or far, and learn a compact embedding that captures distance-dependent acoustic structure. Second, the learned embedding conditions a multichannel U-Net separator via feature-wise linear modulation, providing global spatial context during separation. Experiments are conducted on simulated reverberant mixtures with varying room conditions, array geometries, and near-far source configurations. Results show that the learned embedding forms a structured latent space correlated with source distance and that distance-conditioned separation consistently outperforms unconditioned and handcrafted-feature baselines. In particular, distance-aware conditioning yields approximately 2–3 dB improvements in SI-SDR in near-far separation scenarios, demonstrating the benefit of explicitly modeling distance in multichannel source separation.

****POMA Student Paper Competition Winner***

1. INTRODUCTION

Accurate separation of overlapping sound sources in reverberant environments remains a fundamental challenge in acoustics, with important implications for spatial hearing devices, hearing aids, teleconferencing systems, and acoustic scene analysis. In real-world environments, source signals undergo complex propagation effects, including distance-dependent attenuation, spectral coloration, and an increasing contribution of reverberant energy. These effects substantially alter the spatial cues commonly exploited for multichannel source separation, such as interchannel time and level differences, as well as distance-related measures like the direct-to-reverberant ratio (DRR).

Most existing multichannel source separation methods primarily rely on interchannel spatial diversity, while rarely modeling distance-dependent acoustic structure explicitly. In highly reverberant rooms, source-to-array distance acts as a global attribute that systematically shapes signal energy, spatial coherence, and direct-to-reverberant balance across microphones, often reducing the reliability of conventional narrowband spatial features. Reverberation further smears temporal structure, limiting the effectiveness of approaches that implicitly assume anechoic or mildly reverberant conditions. Consequently, separation performance typically degrades as reverberation time increases and distance disparities between sources grow.

In this work, we investigate a distance-aware source separation framework that treats source-to-array distance not as a target quantity to be estimated, but as a latent global variable governing multichannel spatial statistics. Rather than predicting physical distance explicitly, we learn a compact embedding from multichannel spectral features that captures distance-dependent acoustic structure arising from attenuation, reverberation, and spatial coherence. This learned embedding is used to condition a multichannel separator, enabling it to adapt its internal representations based on the overall near–far spatial configuration of the mixture. We evaluate the proposed approach on simulated reverberant mixtures spanning a range of room sizes, source distances, and array geometries, and show that distance-conditioned separation consistently outperforms unconditioned baselines. These results indicate that explicitly modeling distance-related spatial structure as a latent contextual variable can lead to improved multichannel source separation in realistic reverberant environments.

2. RELATED WORK

Multichannel audio source separation has a long history in statistical signal processing, where spatial diversity across microphones is exploited to disentangle overlapping sources. Early model-based approaches relied on probabilistic formulations that combine spectral structure with spatial covariance modeling, including multichannel nonnegative matrix factorization (MNMF),¹ full-rank spatial covariance models for underdetermined reverberant mixtures,² and Gaussian-model-based approaches incorporating spatial location priors.³ Subsequent work introduced more robust spatial statistics such as the complex angular central Gaussian mixture model (CACGMM),⁴ along with efficient inference based on joint diagonalization.^{5,6} Related efforts explored alternative spatial cues such as interchannel loudness vector sums in multichannel music separation.⁷ More recently, deep learning approaches have advanced multichannel separation by learning joint spectral–spatial representations directly from multichannel STFT inputs, with multichannel U-Net variants often outperforming classical methods when trained on large simulated datasets.^{8,9} However, most learning-based systems still treat spatial cues as relative, local descriptors (e.g., IPD/ILD) rather than as manifestations of a global physical attribute—such as source distance—that systematically shapes multichannel spatial statistics in reverberant environments.

Distance-based sound separation (DSS) has been studied across work on near–far grouping and reverberation-aware processing. Patterson et al.¹⁰ framed DSS as a near–far separation problem relative to a fixed distance threshold, demonstrating that distance-dependent acoustic structure can support separation without explicit dereverberation or RIR estimation, with reported gains of approximately 4–7 dB SI-SNR in multi-speaker scenes. A patent extension emphasized efficient deployment and robustness across environments.¹¹ Related formulations also appear in target speech extraction (TSE), where distance- or room-related information is used as auxiliary conditioning to extract a desired speaker.^{12,13} Despite these advances, existing DSS methods are largely restricted to single-channel or single-microphone settings and typically rely on binary near–far labeling. More broadly, distance is often treated as a supervision signal for output grouping rather than as a latent variable governing the global spatial structure of multichannel observations.

Sound distance estimation from audio has also been studied extensively in psychoacoustics, room acoustics, and robot audition. Foundational studies show that distance perception depends on interacting cues such as overall level, spectral coloration, and the balance between direct and reverberant energy.^{14–16} These findings motivated

signal-processing methods that estimate distance-related quantities using energy decay and DRR cues,^{17,18} as well as learning-based approaches for estimating reverberation-related parameters.¹⁹ While promising, most prior work treats distance estimation as an end goal rather than as contextual information for improving separation.

In contrast, the present work treats source-to-array distance as a latent global attribute that governs the joint spatial statistics of multichannel audio, rather than as an explicit quantity to be estimated or a label for output grouping. We learn a compact distance embedding from multichannel spatial features and condition a separator network on this embedding, enabling adaptation to the global near–far configuration of sources in reverberant environments.

3. PROBLEM FORMULATION AND DISTANCE-DEPENDENT ACOUSTIC CUES

We consider a multichannel acoustic mixture recorded by an array of M microphones in a reverberant enclosure. Let $s_i(t)$ denote the i th source signal and $x_j(t)$ the signal at microphone j . Acoustic propagation from source i to microphone j is modeled by an impulse response $h_{ji}(t)$ that depends on the source-to-array distance d_i , room geometry, and array configuration. The observed mixture is

$$x_j(t) = \sum_{i=1}^I (h_{ji} * s_i)(t) + n_j(t), \quad (1)$$

where $*$ denotes convolution and $n_j(t)$ is additive noise. Under the narrowband approximation, the corresponding STFT-domain model is

$$X_j(f, t) = \sum_{i=1}^I H_{ji}(f) S_i(f, t) + N_j(f, t), \quad (2)$$

where (f, t) index frequency and time frames. In reverberant environments, $H_{ji}(f)$ is complex-valued and frequency-dependent, inducing nontrivial spatial structure and reduced interchannel coherence.

Separation objective. Given the multichannel mixture STFT $\mathbf{X} = \{X_j(f, t)\}_{j=1}^M$, our goal is to estimate the source signals as captured at a reference microphone (i.e., the reference-channel spatial images), consistent with mask-based separation. Throughout, we focus on $I = 2$ (near–far) to isolate distance-related structure; extension to more sources is discussed in Sec. 7.

Distance affects propagation through coupled changes in level, coloration, temporal smearing, and spatial coherence. We summarize this by decomposing the impulse response into direct and reverberant components,

$$h_{ji}(t; d_i) = h_{ji}^{\text{dir}}(t; d_i) + h_{ji}^{\text{rev}}(t; d_i), \quad (3)$$

and by the associated direct-to-reverberant ratio (DRR),

$$\text{DRR}_{ji} = 10 \log_{10} \frac{\int |h_{ji}^{\text{dir}}(t; d_i)|^2 dt}{\int |h_{ji}^{\text{rev}}(t; d_i)|^2 dt}, \quad (4)$$

which typically decreases with increasing d_i . In our implementation, DRR-based proxies are computed via an early–late energy ratio from the acoustic impulse responses, approximating direct versus reverberant energy without explicit decomposition.

These distance-dependent effects also manifest in spatial cues derived from multichannel STFTs. For microphones j_1 and j_2 , we compute interchannel level difference (ILD)

$$\text{ILD}_{j_1, j_2}(f, t) = 20 \log_{10} \frac{|X_{j_1}(f, t)|}{|X_{j_2}(f, t)|}, \quad (5)$$

interchannel phase difference (IPD)

$$\text{IPD}_{j_1, j_2}(f, t) = \angle X_{j_1}(f, t) - \angle X_{j_2}(f, t), \quad (6)$$

and interchannel spatial coherence

$$\gamma_{j_1, j_2}(f) = \frac{\mathbb{E}_t[X_{j_1}(f, t)X_{j_2}^*(f, t)]}{\sqrt{\mathbb{E}_t[|X_{j_1}(f, t)|^2]\mathbb{E}_t[|X_{j_2}(f, t)|^2]}}. \quad (7)$$

As d_i increases and the sound field becomes more diffuse, consistent ILD/IPD patterns and coherence typically weaken, reflecting that these cues depend on global source–array distance and reverberation rather than purely local time–frequency behavior.

Rather than relying on heuristic combinations of Eqs. (5)–(7), we treat source-to-array distance as a latent variable governing the joint distribution of multichannel spatial features. We learn a low-dimensional embedding $\mathbf{z}_i \in \mathbb{R}^D$ from source-specific multichannel observations,

$$\mathbf{z}_i = f_{\text{dist}}(\mathbf{X}^{(i)}), \quad (8)$$

where $\mathbf{X}^{(i)}$ denotes the multichannel STFT of source i alone (its spatial image over the array) and $f_{\text{dist}}(\cdot)$ is trained with distance-related supervision (Sec. 6). In practice, $\mathbf{X}^{(i)}$ is available during training via ground-truth single-source rendering and is used as oracle conditioning in our controlled separation experiments. The embedding serves (i) as an auxiliary representation supporting distance regression and near–far classification during representation learning, and (ii) as a conditioning variable that modulates a multichannel separator.

Given a mixture $\mathbf{X}$, we define a distance-conditioned separator

$$\hat{S}_i = g_{\text{sep}}(\mathbf{X}; \mathbf{z}_1, \mathbf{z}_2), \quad (9)$$

implemented as a complex-valued mask estimator applied to the mixture STFT at a fixed reference channel so that $\hat{S}_i$ corresponds to the estimated reference-channel spatial image. This formulation decouples learning distance-dependent representations from separation, enabling the separator to adapt based on the relative near–far configuration in reverberant environments.

4. DISTANCE EMBEDDING NETWORK

The goal of the distance embedding network (DistanceNet) as shown in Fig. 1 is to learn a compact, low-dimensional representation that captures distance-dependent acoustic structure from multichannel audio. Rather than treating distance estimation as an end goal, DistanceNet is designed to induce a structured latent space that reflects global spatial characteristics associated with source-to-array distance. Given source-specific multichannel observations available during training, DistanceNet produces a normalized embedding that summarizes distance-related structure and is subsequently used to condition the source separation network described in Sec. 5. Distance regression and near–far classification are used only as auxiliary supervision to shape the learned representation.

A. INPUT REPRESENTATION

Let $\mathbf{X}^{(i)} \in \mathbb{C}^{M \times F \times T}$ denote the multichannel STFT corresponding to the *spatial image* of source i over the microphone array (i.e., the source convolved with the array’s acoustic impulse responses). During training, $\mathbf{X}^{(i)}$ is obtained from ground-truth single-source renderings and serves as oracle input for representation learning.

The real and imaginary components are stacked to form a real-valued tensor,

$$\tilde{\mathbf{X}}^{(i)} = [\Re(\mathbf{X}^{(i)}); \Im(\mathbf{X}^{(i)})] \in \mathbb{R}^{2M \times F \times T}. \quad (10)$$

Here, the factor of $2M$ arises from stacking real and imaginary components across microphones. This representation preserves interchannel phase relationships while remaining compatible with real-valued convolutional networks. In some experiments, $\tilde{\mathbf{X}}^{(i)}$ may be augmented with derived spatial features such as ILD and IPD; however, the primary focus of this work is on learning distance-related structure directly from multichannel STFTs.

B. NETWORK ARCHITECTURE

DistanceNet is implemented as a convolutional encoder that aggregates spectro-temporal and spatial information across channels and frequencies. The network consists of a sequence of convolutional blocks, each comprising convolution, batch normalization, and rectified linear unit (ReLU) activation, followed by strided downsampling along the time and frequency dimensions.

After the final encoder block, global average pooling is applied to obtain a fixed-length representation,

$$\mathbf{h} = \text{GAP}(\tilde{\mathbf{X}}_{\text{enc}}^{(i)}) \in \mathbb{R}^C, \quad (11)$$

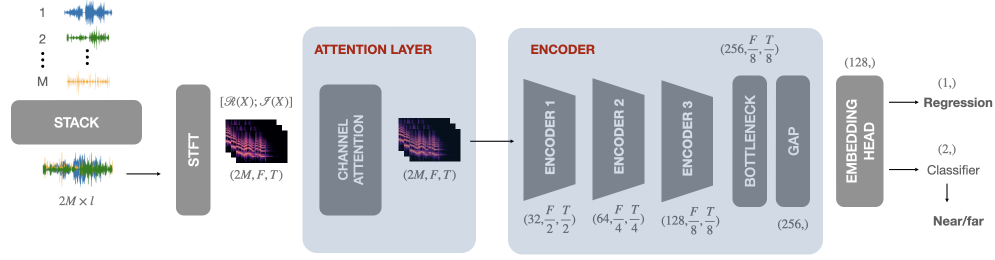


Figure 1: Architecture of the proposed DistanceNet. Source-specific multichannel STFT inputs are encoded using a convolutional encoder, followed by global pooling and three output heads. Distance regression and near–far classification provide auxiliary supervision to induce a structured distance embedding, which is used for conditioning in the separation stage.

where $\tilde{\mathbf{X}}_{\text{enc}}^{(i)}$ denotes the output of the last encoder layer and C is the number of feature channels. This pooled representation is passed to three task-specific output heads:

1. a distance regression head producing $\hat{d} \in \mathbb{R}$,
2. a near–far classification head producing $\hat{\mathbf{y}} \in \mathbb{R}^2$,
3. an embedding head producing $\mathbf{z} \in \mathbb{R}^D$.

The embedding vector is ℓ_2 -normalized,

$$\mathbf{z} \leftarrow \frac{\mathbf{z}}{\|\mathbf{z}\|_2}, \quad (12)$$

ensuring scale invariance and facilitating its use as a conditioning variable in downstream separation.

C. LEARNING OBJECTIVES

DistanceNet is trained using a multi-task objective that combines distance regression and near–far classification to shape the learned embedding space. Let d denote the ground-truth source-to-array distance and $\mathbf{y} \in \{0, 1\}^2$ denote the corresponding near–far label (with $\mathbf{y} = [1, 0]$ for near and $\mathbf{y} = [0, 1]$ for far). The distance regression loss is defined as

$$\mathcal{L}_{\text{dist}} = \|d - \hat{d}\|_1, \quad (13)$$

which is chosen for robustness to room-dependent variability and outliers.

The near–far classification loss is defined using categorical cross-entropy. Let $\hat{\mathbf{y}} \in \mathbb{R}^2$ denote the predicted class probability vector obtained by applying a softmax to the classification head output, and let $\mathbf{y} \in \{0, 1\}^2$ denote the corresponding one-hot ground-truth label. The classification loss is then given by

$$\mathcal{L}_{\text{cls}} = - \sum_{k=1}^2 y_k \log \hat{y}_k. \quad (14)$$

The overall DistanceNet loss is

$$\mathcal{L}_{\text{DN}} = \lambda_{\text{dist}} \mathcal{L}_{\text{dist}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}, \quad (15)$$

where λ_{dist} and λ_{cls} control the relative weighting of each term. We emphasize that these losses regularize the embedding space during representation learning; distance prediction itself is not required during separation.

D. INTERPRETATION OF THE DISTANCE EMBEDDING

Figure 2 illustrates the structure of the learned distance embedding. The embedding $\mathbf{z}$ captures global distance-dependent acoustic structure that is difficult to encode using local time–frequency cues alone. Empirically, embeddings corresponding to near and far sources form well-separated clusters, with smooth transitions as distance varies, indicating that the network learns a continuous organization rather than a purely binary grouping.

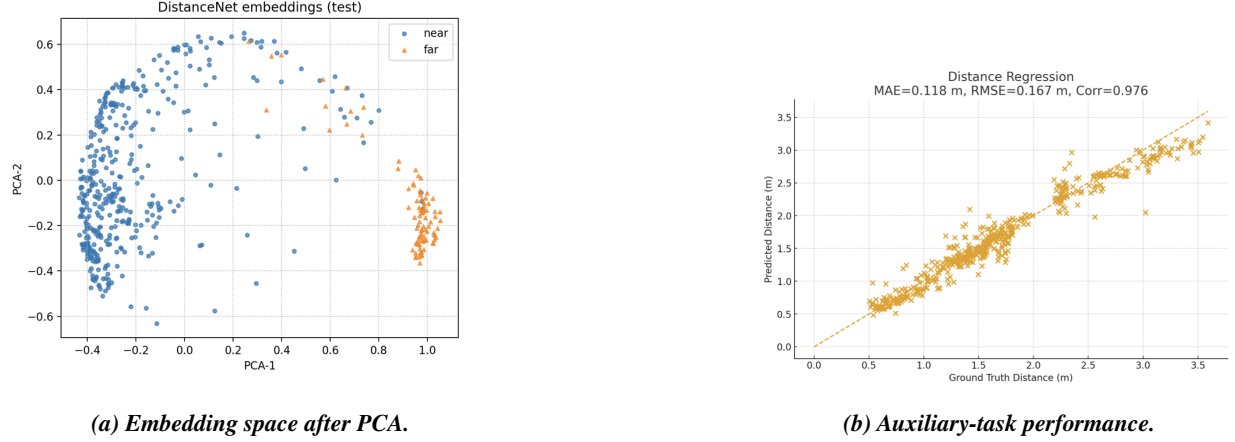


Figure 2: Learned distance embedding structure and auxiliary-task behavior. *Left: two-dimensional principal component analysis (PCA) visualization of the learned embedding, showing smooth organization with respect to distance. Right: distance regression and near–far classification are reported only to verify that the embedding is consistent with distance-related supervision, not to position DistanceNet as a distance-estimation system.*

To visualize this structure, we apply principal component analysis (PCA) to project embeddings into two dimensions. This organization supports the interpretation of distance as a latent global attribute that shapes multichannel spatial statistics. As a result, $\mathbf{z}$ serves as an effective conditioning signal for downstream separation, allowing the separator to adapt its internal representations according to the overall near–far spatial configuration of sources.

5. DISTANCE-CONDITIONED SOURCE SEPARATION

This section describes the proposed distance-conditioned source separation framework. The central idea is to use learned distance embeddings produced by DistanceNet (Sec. 4) as global contextual information that modulates a multichannel separator. Conditioning is implemented using feature-wise linear modulation (FiLM), which allows distance-related context to influence intermediate feature representations throughout the network. In order to isolate the effect of distance conditioning itself, we assume oracle access to source-specific distance embeddings during separation; we do not address mixture-only inference of these embeddings in this work.

A. SEPARATION ARCHITECTURE OVERVIEW

Let $\mathbf{X} \in \mathbb{C}^{M \times F \times T}$ denote the multichannel mixture STFT. As in Sec. 4, the real and imaginary components are stacked to form

$$\tilde{\mathbf{X}} = [\Re(\mathbf{X}); \Im(\mathbf{X})] \in \mathbb{R}^{2M \times F \times T}. \quad (16)$$

The separator is implemented as a multichannel U-Net that estimates complex-valued masks for each source. The U-Net follows an encoder–decoder structure with skip connections, where encoder blocks progressively aggregate spectro-temporal context and decoder blocks reconstruct source-specific representations.

Unlike conventional multichannel separators, the proposed architecture incorporates distance conditioning at multiple depths via FiLM, allowing global distance-related context to modulate both low-level spatial feature extraction and higher-level source reconstruction. Fig. 3 provides an overview of the proposed distance-conditioned U-Net separator.

B. FEATURE-WISE LINEAR MODULATION (FiLM)

Feature-wise linear modulation (FiLM)²⁰ is a conditioning mechanism that modulates neural network activations using auxiliary information. Rather than concatenating conditioning variables with network inputs, FiLM applies an affine transformation to intermediate feature maps, allowing conditioning information to influence the network in a structured and interpretable manner.

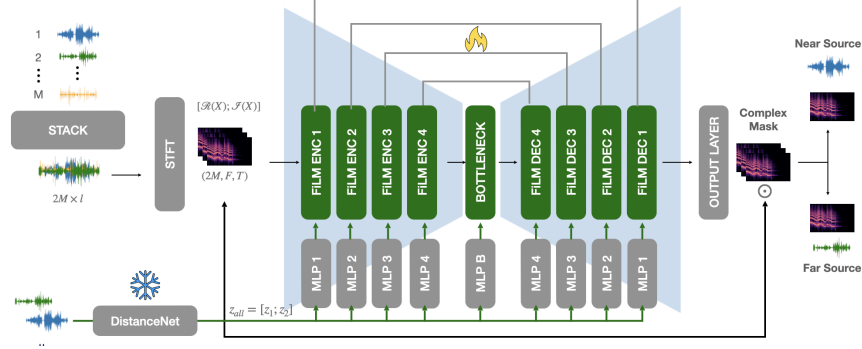


Figure 3: Overview of the distance-conditioned U-Net separator. Source-specific distance embeddings are used to modulate encoder and decoder feature maps via FiLM layers, injecting global distance context throughout the network.

Let $\mathbf{H}_\ell \in \mathbb{R}^{C_\ell \times F_\ell \times T_\ell}$ denote the activation tensor at layer ℓ of the separator. A FiLM layer transforms $\mathbf{H}_\ell$ as

$$\text{FiLM}(\mathbf{H}_\ell \mid \gamma_\ell, \beta_\ell) = \gamma_\ell \odot \mathbf{H}_\ell + \beta_\ell, \quad (17)$$

where $\gamma_\ell, \beta_\ell \in \mathbb{R}^{C_\ell}$ are learned scale and bias parameters, respectively, and $\odot$ denotes channel-wise multiplication with broadcasting across time and frequency.

The scale parameters γ_ℓ control the relative emphasis of feature channels, while the bias parameters β_ℓ shift activation thresholds. This enables the network to selectively emphasize or suppress spectro-spatial patterns based on global conditioning information.

C. DISTANCE-BASED FiLM CONDITIONING

In the proposed framework, FiLM parameters are derived from the distance embeddings associated with the sources. Let $\mathbf{z}_1, \mathbf{z}_2 \in \mathbb{R}^D$ denote the normalized embeddings corresponding to the near and far sources, respectively. These embeddings are concatenated to form a global conditioning vector,

$$\mathbf{z}_{\text{all}} = [\mathbf{z}_1; \mathbf{z}_2] \in \mathbb{R}^{2D}. \quad (18)$$

For each U-Net layer ℓ , an independent multilayer perceptron (MLP) maps $\mathbf{z}_{\text{all}}$ to FiLM parameters,

$$(\gamma_\ell, \beta_\ell) = \text{MLP}_\ell(\mathbf{z}_{\text{all}}), \quad (19)$$

where $\gamma_\ell, \beta_\ell \in \mathbb{R}^{C_\ell}$. This design allows different layers to respond differently to distance-related context, with earlier layers adapting low-level spatial cues and deeper layers modulating higher-level source representations.

In this work, $\mathbf{z}_1$ and $\mathbf{z}_2$ are computed from ground-truth single-source spatial images and provided as oracle conditioning during separation. This controlled setting isolates the contribution of distance-aware conditioning independently of errors in distance inference.

D. INTERPRETATION FROM AN ACOUSTIC PERSPECTIVE

From an acoustic perspective, source distance influences signal level, spectral tilt, direct-to-reverberant ratio (DRR), and spatial coherence across microphones. These effects are global properties of a source relative to the array and are difficult to infer reliably from isolated time–frequency bins.

FiLM enables the separator to incorporate this global distance-related context by reshaping its internal feature space. For example, when a source is far from the array, FiLM may suppress features associated with strong inter-channel coherence and emphasize representations that are robust to diffuse reverberation. Conversely, for near sources, FiLM may prioritize coherent, high-DRR features. This behavior aligns with classical acoustic intuition while remaining fully data-driven.

E. MASK ESTIMATION AND RECONSTRUCTION

The output of the U-Net consists of complex-valued masks $M_i(f, t)$ for each source i , represented by predicted real and imaginary mask components. Estimated source STFTs are obtained as

$$\hat{S}_i(f, t) = M_i(f, t) \odot X_{\text{ref}}(f, t), \quad (20)$$

where $X_{\text{ref}}(f, t)$ denotes the STFT of a fixed reference microphone channel (we use $X_{\text{ref}} = X_1$ throughout). This choice yields estimates of the source spatial images at the reference microphone. Time-domain signals are reconstructed using inverse STFT with overlap-add.

F. SEPARATION LOSS

The separator is trained using a scale-invariant signal-to-distortion ratio (SI-SDR) loss computed in the time domain,

$$\mathcal{L}_{\text{sep}} = - \sum_{i=1}^2 \text{SI-SDR}(s_i, \hat{s}_i), \quad (21)$$

where $s_i(t)$ and $\hat{s}_i(t)$ denote the ground-truth and estimated reference-channel source signals, respectively. Since outputs are ordered as (near, far) via oracle conditioning, permutation-invariant training is not required.

The overall system is trained sequentially: DistanceNet is trained first and its parameters are fixed while training the distance-conditioned separator.

6. EXPERIMENTAL SETUP

This section describes the dataset generation procedure, acoustic simulation parameters, training protocol, baseline configurations, and evaluation metrics used to assess the proposed distance-aware source separation framework.

A. DATASET GENERATION

Obtaining real multichannel recordings with precise source-to-array distance annotations is challenging, particularly in reverberant environments. To enable controlled and repeatable evaluation, experiments are therefore conducted using simulated reverberant mixtures. Multichannel acoustic impulse responses and microphone signals are generated using *pyroomacoustics*²¹ in a shoebox room via the image-source method with Sabine-based absorption calibration, allowing systematic variation of source distance, room characteristics, and microphone array geometry while preserving physically meaningful propagation effects.

Each mixture consists of two speech sources positioned at different distances relative to the microphone array (near and far). Source signals are drawn from LibriSpeech²² and rendered through simulated rooms. In the released generator, each source waveform is RMS-normalized and scaled to a fixed level prior to propagation, and the final mixture is peak-normalized to avoid clipping. Unless otherwise noted, no additional additive noise beyond that implicit in the simulated acoustic impulse responses is included.

B. ROOM AND ARRAY CONFIGURATIONS

Room dimensions are randomly sampled with length $L \in [3, 10]$ m, width $W \in [3, 8]$ m, and height $H \in [2.5, 4]$ m, with reverberation times drawn from $\text{RT}_{60} \in \{0.2, 0.4, 0.6, 0.8\}$ s to span mild to moderately reverberant conditions representative of small and medium-sized enclosed spaces.

Robustness is evaluated across multiple microphone array geometries (linear, circular, and spherical) using $M = 8$ microphones, reflecting a practical trade-off between spatial diversity and compact array size. The array is centered in the room at a height of 1.3 m, with geometry-specific parameters of 4 cm linear spacing, 8 cm circular radius, and 10 cm spherical radius using Fibonacci-sphere placement. Sources are positioned at a height of 1.5 m with a wall-clearance margin of 0.3 m using a geometry-aware sampler to ensure physically feasible placements. The spherical array configuration follows standard practices in spatial audio and array processing.²³

Source-to-array distances are sampled from predefined ranges, with near sources drawn from $[0.5, 2.0]$ m and far sources from $[3.0, 6.0]$ m, subject to room-dependent feasibility constraints. Near-far labels are assigned using a distance threshold of 2.5 m, providing a clear separation between distance regimes while avoiding ambiguous boundary cases.

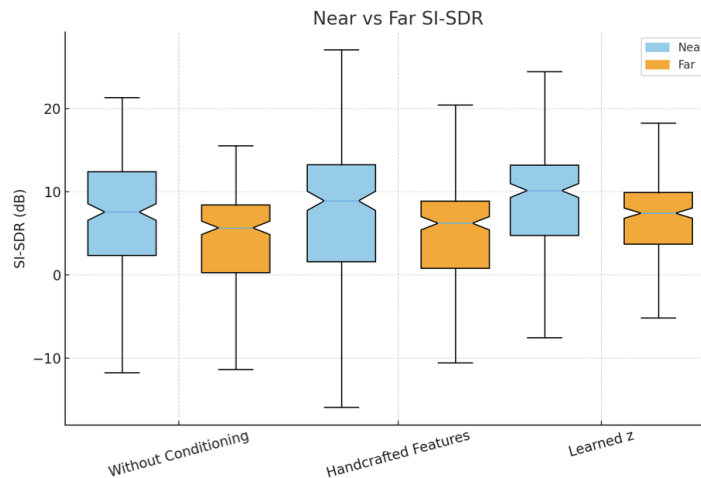


Figure 4: Near and far source separation performance (SI-SDR) for unconditioned, handcrafted-feature-conditioned, and distance-embedding-conditioned models.

C. TRAINING PROCEDURE

DistanceNet and the distance-conditioned separator are trained sequentially. DistanceNet is trained using single-source multichannel spatial images with known source-to-array distances and near-far labels, minimizing the multi-task loss defined in Sec. 4.

After DistanceNet training, its parameters are fixed. For the separation experiments reported here, we provide *oracle distance embeddings* computed from ground-truth single-source spatial images for each mixture. These embeddings are concatenated and used to condition the separator via FiLM layers as described in Sec. 5, isolating the effect of distance-aware conditioning independently of errors in distance inference. The separator is trained using the SI-SDR loss in Sec. 5, with early stopping based on validation performance.

D. BASELINES

To isolate the contribution of distance-aware conditioning, we evaluate three baselines: (i) an *unconditioned separator*, implemented as a multichannel U-Net trained without any explicit distance or spatial conditioning; (ii) *handcrafted feature conditioning*, where the separator is conditioned on explicit spatial features including ILD, IPD, and DRR-based proxies; and (iii) *learned distance embedding*, corresponding to the proposed DistanceNet-based FiLM conditioning approach. All models share the same backbone architecture, optimization procedure, and training data to ensure a fair comparison. For the unconditioned baseline, source ordering is determined by energy-based assignment to near and far reference channels; permutation-invariant training is not used in order to match the fixed ordering induced by oracle distance conditioning in the proposed method.

E. EVALUATION METRICS

Separation performance is evaluated using scale-invariant signal-to-distortion ratio (SI-SDR), computed separately for near and far sources at the reference microphone. We report median SI-SDR across the test set to reduce sensitivity to outliers. In addition, we visualize SI-SDR distributions using box plots and qualitatively analyze the learned distance embeddings to assess their structure and consistency.

7. RESULTS AND DISCUSSION

This section presents quantitative and qualitative results evaluating the proposed distance-aware source separation framework. We focus on three questions: (i) whether incorporating distance-related context improves separation

Table 1: Median SI-SDR (dB) for near and far sources across different conditioning strategies. Mean denotes the average of near and far SI-SDR.

System	Near $\uparrow$	Far $\uparrow$	Mean $\uparrow$
Unconditioned separator	7.6	5.7	6.65
Handcrafted spatial features	8.9	6.2	7.55
Learned distance embedding	10.1	7.4	8.75

performance, (ii) whether learned distance embeddings provide advantages over handcrafted spatial features, and (iii) how distance conditioning affects near and far sources differently.

A. DISTANCE EMBEDDING CONSISTENCY

DistanceNet exhibits stable behavior on its auxiliary training tasks, including near–far classification and distance regression. Near–far classification accuracy exceeds 93% across all evaluated array geometries, indicating that distance-related acoustic structure is consistently captured from multichannel STFT inputs.

Figure 2 illustrates properties of the learned distance embedding. Distance regression outputs show a strong correspondence with ground-truth distances (mean absolute error of 0.118 m, root mean square error of 0.167 m, and Pearson correlation coefficient of 0.976). These results are reported to verify that the learned embedding varies smoothly with distance and reflects continuous distance-related structure, rather than to position DistanceNet as a distance estimation system.

Qualitative inspection of the embedding space (Fig. 2, left) reveals clear organization with respect to near and far source configurations, with gradual transitions as distance increases. This structure emerges without explicitly enforcing metric constraints during training, supporting the interpretation of distance as a latent global attribute shaping multichannel spatial statistics in reverberant environments.

B. SOURCE SEPARATION PERFORMANCE

Table 1 summarizes separation performance in terms of SI-SDR for near and far sources. Conditioning the separator on distance-related context yields consistent improvements over the unconditioned baseline across all evaluated configurations. Fig. 4 visualizes the distribution of SI-SDR values for near and far sources across conditioning strategies.

Near-source performance benefits most strongly from distance-aware conditioning. Compared to the unconditioned separator, handcrafted spatial features provide modest improvements, while learned distance embeddings yield gains of approximately 2–3 dB SI-SDR. This indicates that the learned embedding captures distance-dependent structure beyond what can be represented using explicit ILD, IPD, or DRR-based proxies alone.

Far-source separation also improves with distance conditioning, although gains are smaller in magnitude. This behavior is expected, as far sources are more heavily influenced by reverberation and exhibit reduced interchannel coherence. Nonetheless, the learned embedding consistently outperforms handcrafted spatial features, indicating that distance-aware conditioning remains beneficial even under more challenging reverberant conditions.

The observed improvements highlight the effectiveness of feature-wise linear modulation as a mechanism for incorporating global distance-related context into a multichannel separator. Unlike input-level conditioning, FiLM allows distance information to influence intermediate feature representations at multiple depths of the network.

From an acoustic perspective, this enables the separator to adapt its internal processing based on expected spatial characteristics. For near sources, FiLM emphasizes coherent, high-DRR features and sharper temporal structure. For far sources, FiLM suppresses unreliable narrowband phase cues and promotes representations that are more robust to diffuse reverberation. These behaviors align with classical acoustic intuition and emerge naturally through data-driven training.

A consistent observation across experiments is that learned distance embeddings outperform handcrafted spatial features. While ILD, IPD, and DRR-based proxies provide useful localized information, they are inherently sensitive to reverberation and temporal variability. In contrast, the learned embedding integrates spatial information across time, frequency, and channels, yielding a more stable and globally consistent representation of distance-related structure.

These results suggest that distance-aware source separation benefits from learning spatial representations directly from data rather than relying solely on hand-engineered cues derived from simplified acoustic assumptions.

Directional cues and distance confounds. We note that both handcrafted spatial features and learned embeddings may implicitly encode directional information in addition to distance-related cues. However, because all methods are trained and evaluated on mixtures drawn from the same joint distribution over array geometries, room conditions, and source configurations, relative performance differences can be attributed to the conditioning mechanism rather than to geometry-specific biases.

C. LIMITATIONS

The experiments presented here rely on simulated mixtures with known source distances and two-source near-far configurations. Although simulation enables controlled evaluation, real-world recordings may exhibit additional complexities such as source motion, nonstationary background noise, and microphone mismatch. Furthermore, the current framework assumes access to source-specific distance embeddings during separation in order to isolate the effect of distance-aware conditioning. Developing mixture-only distance inference and joint learning strategies remains an important direction for future work.

8. CONCLUSION

This paper examined the role of distance-dependent acoustic structure in multichannel source separation under reverberant conditions. While many separation methods treat sources primarily as abstract signals, our results show that source-to-array distance functions as a global attribute that systematically shapes signal level, reverberation, and spatial coherence across microphones.

We proposed a distance-aware framework consisting of a distance embedding network (DistanceNet) and a FiLM-conditioned multichannel separator. DistanceNet learns a compact representation of distance-related spatial structure from multichannel STFT features, while FiLM conditioning enables the separator to adapt its internal representations according to the relative near-far configuration of sources. Experiments on simulated reverberant mixtures demonstrated that incorporating distance-aware conditioning leads to consistent separation improvements compared to unconditioned and handcrafted-feature-based baselines, with gains of approximately 2–3 dB SI-SDR in near-far scenarios.

Beyond quantitative improvements, the proposed formulation offers a physically grounded perspective on multichannel source separation by explicitly linking learned representations to distance-dependent acoustic effects. In this work, oracle access to distance embeddings was assumed in order to isolate the impact of distance-aware conditioning. Extending this framework to mixture-only distance inference, joint learning of distance representations and separation, and more complex spatial configurations involving additional sources remains an important direction for future investigation.

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